Luwei Wang

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Profile

PhD candidate in Biomedical AI at the University of Edinburgh, focusing on interpretable and robust machine learning methods for personalised medicine and clinical decision support. My research lies at the intersection of machine learning, causal inference, and biomedicine, where I collaborate with domain experts to develop flexible, scalable, and practically actionable AI solutions for real-world healthcare data.

Education

University of Edinburgh – PhD in Biomedical AI 2023 – 2027 (Expected)

- Research focus: Bayesian modelling, clustering, semi-supervised learning, causal ML, deep learning

University of Edinburgh – MSc in Statistics with Data Science (Distinction) 2021 – 2022

Hong Kong Baptist University – BSc (Hons) Mathematics and Statistics 2017 – 2021
Minor in Computer Science (First Class)

Publications

- **Wang, L.**, Richards, K., Seth, S. *Nonparametric Bayesian Multi-facet Clustering for Longitudinal Data*. Proceedings of the 41st Conference on Uncertainty in Artificial Intelligence (UAI), 2025.
- **Wang, L.**, Lone, N., Seth, S. *Bayesian Supervised Causal Clustering for Heterogeneous Treatment Effects*. Under review.

Research Experience

Research Assistant Oct 2022 – Jan 2023
Data Science Unit, School of Informatics, University of Edinburgh United Kingdom

- Applied supervised clustering methods (Supervised K-means, Mahalanobis metric learning, VAE-based models) to protein biomarker data from critically ill COVID-19 patients.
- Identified outcome-guided patient subphenotypes to study heterogeneity of treatment effects and patient stratification for precision treatment strategies.
- Delivered a technical report evaluating the effectiveness and limitations of supervised clustering in clinical settings.

Selected Projects

Bayesian Semi-supervised Clustering with Constraints Ongoing

- Developed a scalable Bayesian semi-supervised clustering model for large-scale EHR data ($\sim$500k patients) with binary disease indicators.
- Implemented stochastic variational inference, reducing runtime by over 80% while jointly learning cluster structure and feature relevance.

Clustering Trajectories of Multiple Long-term Conditions [Website]

- Clustered longitudinal GP records from $\sim$150k UK Biobank participants to identify multimorbidity phenotypes and predict adverse outcomes.
- Proposed a K-means + RNN framework with attention-based temporal feature extraction, yielding more distinct disease-dominant clusters.

Skills

Programming: Python, R, SQL, MATLAB, JAVA

Frameworks & Tools: PyTorch, Stan, LaTeX